Notes: Kaplan & Rauh (JEP, 2013)

It's the Market: The Broad-Based Rise in the Return to Top Talent

Contents

1 Big picture

- **Question.** Why did income at the very top rise so much in the United States? Is the rise mainly a product of managerial power and social norms, or is it a broad market return to scalable talent?
- **Main claim.** The rise in top incomes is broad-based across occupations: public-company CEOs, private-company executives, hedge fund and private equity investors, lawyers, and professional athletes all experienced large gains.
- **Core interpretation.** The evidence is more consistent with a superstar/scale story than with a narrow managerial-power story. Information technology, communications, finance, and globalized markets allow top talent to operate over larger scale.
- **Key contrast.** If CEO pay were mainly a public-company governance problem, the rise should be concentrated in public-company CEOs. The paper argues that it is not.
- **Income evidence.** The top 1 percent income share rose sharply since 1980, but after-tax and after-transfer shares increased less and were closer to late-1990s levels by 2011.
- **Wealth evidence.** The Forbes 400 became less dominated by inherited wealth and more populated by first-generation wealth creators, especially in scalable sectors like technology, finance, and mass retail.
- **US versus non-US.** Similar patterns appear in the non-US billionaire data: fewer inherited fortunes and more first-generation wealth, though the sector composition differs because natural resources and mining play a larger role outside the United States.
- **What the paper is not saying.** The authors do not claim that managerial power, norms, or taxes are irrelevant. Their argument is that the cross-occupation and wealth-origin patterns are harder to reconcile with those explanations as the main driver.
- **Takeaway.** The rise in top income is best understood as a broad increase in the market return to top talent in scalable activities, not merely as a redistribution of rents inside public corporations.

2 Data and measurement

2.1 Occupation-level evidence on top incomes

The paper reviews several groups whose pay is visible enough to compare over time:

- CEOs of S&P 500 firms and other ExecuComp firms.
- Executives in public and private companies using IRS-based classifications from Bakija, Cole, and Heim.
- Top hedge fund managers using *Absolute Return + Alpha*'s annual Rich List.
- Law partners using *American Lawyer* data on profits per partner at large law firms.
- Professional athletes using the USA Today sports salary database.

Interpretation: The point of comparing these occupations is not that they perform identical work. It is that their pay is set in different institutional environments. Broad pay growth across all of them is evidence against a narrow public-company-governance explanation.

2.2 CEO pay measures

The paper distinguishes realized and estimated CEO pay:

$$CEORealizedPay_t = salary_t + bonus_t + restrictedStock_t + optionExerciseValue_t + otherPay_t.$$

Estimated pay instead values options at grant date:

$$CEOEstimatedPay_t = salary_t + bonus_t + restrictedStock_t + optionGrantValue_t + otherPay_t.$$

The authors also scale CEO pay by the average adjusted gross income of the top 0.1 percent:

$$RelativeCEOPay_t = \frac{AverageCEOPay_t}{AverageAGI_{Top\,0.1\%,t}}.$$

Interpretation: Realized pay can be volatile because option exercise values depend on past stock returns. The relative-pay ratio asks whether CEOs rose faster than other top earners, not just whether CEO pay rose in dollars.

2.3 Top income shares

The top income share is:

$$TopShare_{p,t} = \frac{\sum_{i \in Top\,p,t} Income_{i,t}}{\sum_i Income_{i,t}},$$

where p can be the top 1 percent or top 0.1 percent.

The paper compares three versions:

- pre-tax, pre-transfer market income from Piketty and Saez;
- after transfers and before taxes from the Congressional Budget Office;
- after transfers and after taxes from the Congressional Budget Office.

Interpretation: The distinction matters because top income inequality looks more extreme before taxes and transfers than after the fiscal system is applied.

2.4 Forbes 400 and billionaire data

The wealth evidence uses Forbes lists:

- US Forbes 400 in 1982, 1992, 2001, and 2011.
- Forbes non-US billionaires in 1987, 1992, 2001, and 2011.
- For each individual, the authors code the business that created the wealth, the generation of the wealth-creating business, and whether the individual grew up with little/no wealth, some wealth, or wealth.

The generation variable is coded as:

$$Generation = \begin{cases} 1, & \text{founder or first-generation wealth creator,} \\ 1.5, & \text{inherited modest business and built it into a much larger one,} \\ 2, 3, \ldots, & \text{second, third, or later generation family wealth.} \end{cases}$$

Interpretation: The Forbes data are not perfect measures of total wealth inequality. Their value is compositional: they reveal who is at the very top and how those fortunes were created.

2.5 Industry categories

The paper codes wealth-creating businesses into industry categories. The major groups are:

- industrial businesses such as retail/restaurants, computer technology, medical technology, media, energy, and diversified/other;

- finance and investments such as hedge funds, private equity, money management, and venture capital;
- real estate.

Interpretation: The industry coding tests whether top wealth is shifting toward scalable sectors. A technology or finance entrepreneur can deploy skill over a very large market, which is the key Rosen-style superstar mechanism.

3 Framework: competing explanations and empirical tests

3.1 Superstar and scale explanation

The superstar explanation is based on Rosen-style scale economics. Let ability be a_i and the scale over which talent is applied be N_t. The market value of talent can be written schematically as

$$Pay_{i,t} \approx f(a_i) \times N_t.$$

If technology and market integration increase N_t, then the payoff to the very best individuals rises disproportionately.

Interpretation: This mechanism predicts broad increases in top pay across many sectors where skill can be scaled, including executives, finance, lawyers, athletes, and technology entrepreneurs.

3.2 Managerial power explanation

The managerial-power explanation argues that public-company executives obtain pay above marginal product because governance constraints are weak:

$$Pay_{CEO,t} = MP_{CEO,t} + Rent_{governance,t}.$$

Interpretation: If this explanation is the main driver, one would expect the rise in top incomes to be concentrated in public-company CEOs, where dispersed ownership and board capture are most relevant.

3.3 Social norms explanation

The social-norms explanation argues that norms limiting high pay weakened:

$$Pay_{top,t} = MP_{top,t} + Rent_{norms,t}.$$

Interpretation: This explanation predicts that visible and norm-sensitive pay categories, such as public-company CEOs, should show especially large changes. It is less obvious why hidden or private income categories should move similarly.

3.4 Tax-rate bargaining explanation

A tax-based rent-extraction story says lower marginal tax rates increase incentives for top earners to bargain for larger pretax pay because they keep more of the gain:

$$\frac{\partial BargainingEffort}{\partial(1-\tau)} > 0.$$

Interpretation: The paper treats this as possible but incomplete. Technology entrepreneurs and founders of new scalable businesses do not fit easily into a story based mainly on bargaining against lower-wage workers.

3.5 Empirical logic of the paper

The paper's empirical logic is comparative rather than regression-based:

- compare public CEOs to other top earners;
- compare public-company executives to private-company executives;
- compare CEOs to finance professionals, lawyers, and athletes;
- compare income shares before and after taxes/transfers;
- compare inherited versus self-made wealth among the Forbes 400 and non-US billionaires;
- compare industry composition of wealth-creating businesses over time.

Interpretation: The paper's identification is pattern matching across several datasets. The power of the evidence comes from the same broad pattern appearing in many different institutional settings.

4 Identification and credibility

4.1 What makes the evidence informative

- **Cross-occupation breadth.** Top pay rose in many occupations, not just among public-company CEOs.
- **Different governance environments.** Private executives, hedge fund managers, and athletes do not share the same board-capture problem as public-company CEOs.
- **Different visibility.** Some groups have highly visible pay and others do not. Similar increases weaken the pure social-norms interpretation.
- **Wealth-origin changes.** More first-generation fortunes and fewer inherited fortunes point to changing market opportunities rather than simply dynastic accumulation.
- **Sectoral shifts.** Technology and finance categories grew in the Forbes 400, consistent with the return to scalable skill.

Interpretation: The evidence is not a randomized causal design. It is a broad descriptive test of which theory best fits many facts at once.

4.2 Limits and caveats

- The paper does not estimate a structural model of talent or marginal product.
- Forbes lists are imperfect measures of wealth and may miss hidden or offshore wealth.
- Occupational pay data are not perfectly comparable across sectors.
- CEO pay measures depend on how options are valued and when they are realized.
- Broad patterns do not rule out managerial power, social norms, or tax incentives as partial explanations.
- The analysis is primarily about the very top of the distribution, especially the top 1 percent, top 0.1 percent, Forbes 400, and billionaires.

Interpretation: The conclusion should be read as comparative: the authors argue the market/scale explanation fits better than the alternatives, not that it explains every component of inequality.

5 Tables and figures

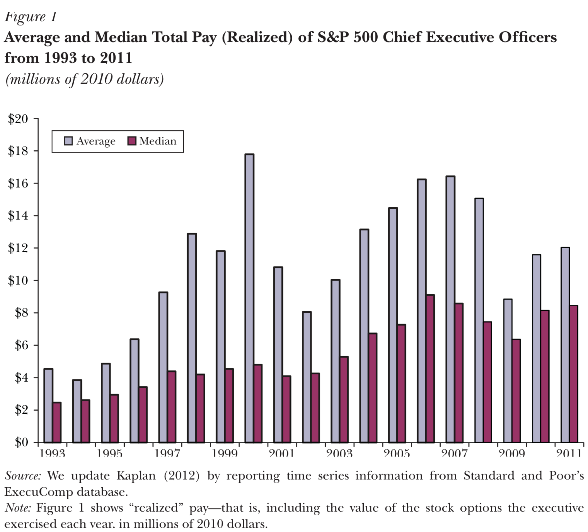
The visuals are grouped to save space and make comparisons faster. All figures and tables below are directly cropped from the paper.

5.1 Figures 1 and 2: CEO pay in dollars and relative to top earners

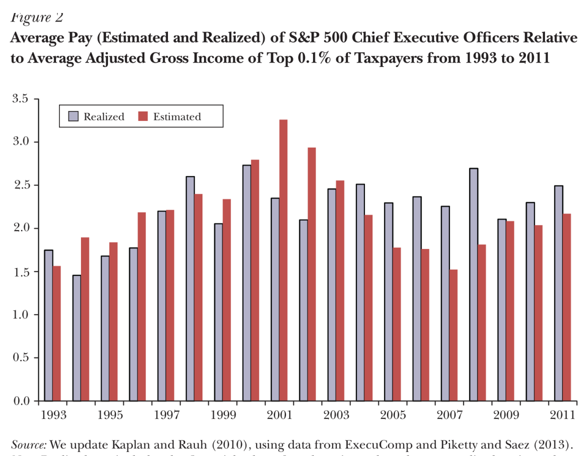
Read:

- Figure 1 shows realized S&P 500 CEO pay rising sharply in the 1990s, peaking around 2000, falling after the dot-com bust, rising again before 2008, then dropping during the financial crisis.
- Average realized CEO pay is much more volatile than median pay because large option exercises by a few CEOs move the mean.
- Median pay trends upward more steadily and peaks around the mid-2000s.
- Figure 2 scales CEO pay by the average adjusted gross income of top 0.1 percent taxpayers.
- Since the late 1990s, S&P 500 CEO pay is generally around 2 to 2.75 times the top 0.1 percent average, rather than rising without bound relative to other top earners.

Economic message: CEO pay rose a lot, but it did not rise uniquely relative to the rest of the top 0.1 percent. This weakens a CEO-specific managerial-power explanation.



(a) Realized pay of S&P 500 CEOs.



(b) CEO pay relative to top 0.1 percent AGI.

5.2 Table 1 and Figure 3: finance, law, and sports

Read:

- Table 1 shows very high pay for the top 25 hedge fund managers. Average pay reaches $882.8 million in 2010, equal to 177.6 times the average AGI of the top 0.1 percent.
- Top law-firm partners also earn high incomes, but at much lower levels than the top hedge fund managers: around $1.56 million per partner in 2010.
- Figure 3 shows top professional athletes also experiencing large increases in pay.
- By 2009, top baseball, basketball, and football salaries are far above their 1993 levels.

Economic message: The rise at the top is not confined to public-company executives. It appears in finance, law, and sports, which are sectors where top talent can be scaled and priced competitively.

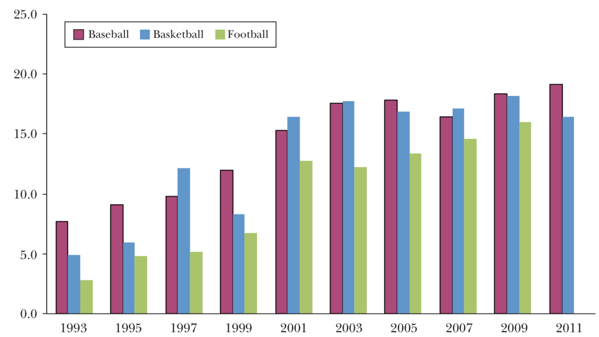
Table 1
Average Pay of Top Hedge Fund Managers and Law Partners

Year	*Average pay of top 25 hedge fund managers* (Millions of $2010 [and relative to average adjusted gross income of top 0.1%])	*Average profit per partner at top 50 law firms* (Millions of $2010 [and relative to average adjusted gross income of top 0.1%])
1994		$0.704 [0.268]
1996		$0.784 [0.219]
1998		$0.997 [0.200]
2000		$1.084 [0.167]
2002	$133.7 [34.6x]	$1.099 [0.285]
2004	$289.5 [55.7x]	$1.286 [0.247]
2006	$616.2 [90.3x]	$1.491 [0.218]
2008	$469.8 [82.1x]	$1.449 [0.253]
2010	$882.8 [177.6x]	$1.557 [0.313]
2012	$537.2 [115.7x]	

Source: Authors using data from *Absolute Return + Alpha* magazine, *American Lawyer*, and Piketty and

(a) Top hedge fund managers and law partners.

Figure 3
Average Top 25 Salaries in Professional Baseball, Basketball, and Football
(in millions of 2010 dollars)



Source: USA Today Sports Salary Database.

(b) Top 25 salaries in professional sports.

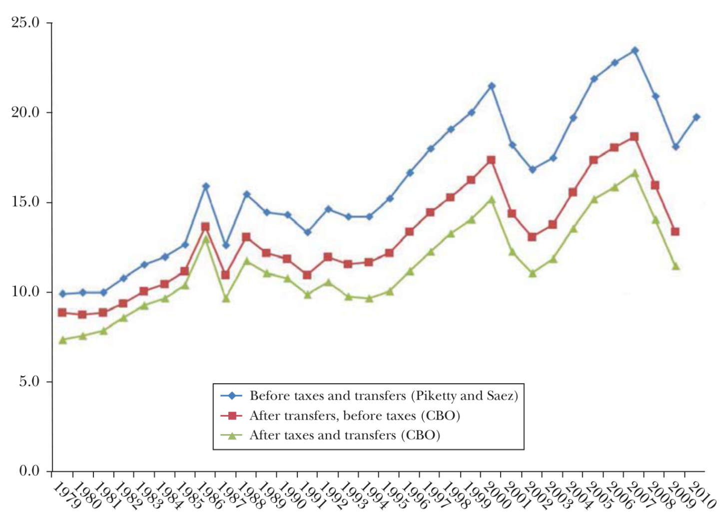
5.3 Figure 4: income share of the top 1 percent

Read:

- The top 1 percent pre-tax, pre-transfer share rises sharply from the late 1970s to 2007.
- The pre-tax share peaks around 23.5 percent in 2007, falls during the crisis, and partially recovers.
- After-transfer and after-tax measures are lower and show a more muted rise.
- The fiscal system therefore dampens the inequality visible in market-income data.

Economic message: Rising market income at the top is real, but taxes and transfers matter for how much of that rise becomes disposable income.

Figure 4
Share of Income Accruing to the Top 1 Percent



Top 1 percent income share before and after taxes/transfers.

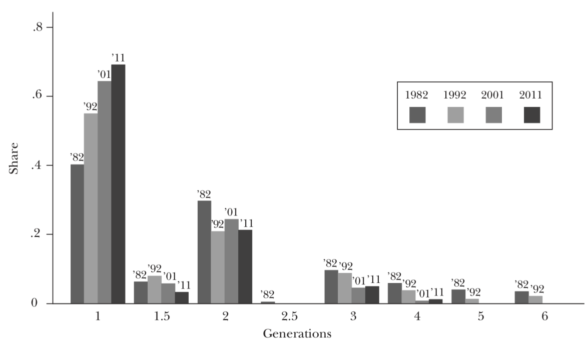
5.4 Figures 5 and 6: US Forbes 400 origins

Read:

- Figure 5 shows a large shift toward first-generation Forbes 400 wealth creators.
- The first-generation share rises from about 40 percent in 1982 to about 69 percent in 2011.
- Later-generation inherited wealth becomes less common.
- Figure 6 shows that the share who grew up wealthy falls from about 60 percent to around 32 percent.
- The "some wealth" category rises, while the little/no wealth category remains roughly flat.

Economic message: The Forbes 400 becomes less inherited and more entrepreneurial. The new rich are often not poor, but they are less likely to come from old dynastic wealth.

Figure 5
Generation of the Wealth-Creating Businesses of Forbes 400 Individuals in 1982, 1992, 2001, and 2011
(share among Forbes 400 individuals)



Source: Authors, using *Who's Who* and Internet searches as primary sources.
Notes: The numbers along the *x*-axis give the generation of a Forbes 400 individual within the family of the founder of his or her business. The normal coding for the generation is an integer. However, if the individual inherited a modest business and built it into a much larger one, we input the generation

(a) Generation of Forbes 400 wealth-creating businesses.

Figure 6
Did the Forbes 400 Grow Up Wealthy?
(share of Forbes 400 individuals for 1982, 1992, 2001, and 2011 with each upbringing)

Source: Authors using *Who's Who* and Internet searches as primary sources.
Notes: Figure 6 shows the share of Forbes 400 individuals for 1982, 1992, 2001, and 2011 who grew up with little or no wealth, who grew up with some wealth, and who grew up wealthy. In coding the data, we view the "some wealth" category as the equivalent of an upper middle class upbringing.

(b) Forbes 400 childhood wealth background.

5.5 Table 2: industries behind the US Forbes 400

Read:

- Retail/restaurant rises from 5.3 percent in 1982 to 15.0 percent in 2011.
- Computer technology rises from 3.3 percent to 12.3 percent.
- Hedge funds rise from 0.5 percent to 7.5 percent.
- Private equity rises from 1.8 percent to 6.8 percent.
- Energy declines from 21.4 percent to 9.8 percent.
- Real estate declines from 17.9 percent to 7.5 percent.

Economic message: US top wealth shifts away from old-style real estate and energy wealth and toward technology, retail scale, and finance.

Table 2
Categories of the Wealth-Creating Businesses behind the Forbes 400
(share of Forbes 400 businesses)

	1982	*1992*	*2001*	*2011*	*Change from 1982 to 2011*
Industrial					
Retail/Restaurant	0.053	0.118	0.132	0.150	+0.097
Technology–Computer	0.033	0.053	0.130	0.123	+0.090
Technology–Medical	0.005	0.018	0.021	0.023	+0.017
Consumer	0.131	0.174	0.125	0.108	−0.023
Media	0.136	0.132	0.164	0.100	−0.036
Diversified/Other	0.207	0.205	0.156	0.123	−0.084
Energy	0.214	0.089	0.062	0.098	−0.117
Finance and investments					
Hedge funds	0.005	0.011	0.018	0.075	+0.070
Private equity/leveraged buyout	0.018	0.034	0.039	0.068	+0.050
Money management	0.018	0.055	0.062	0.045	+0.027
Venture capital	0.003	0.005	0.008	0.015	+0.012
Real estate	0.179	0.105	0.081	0.075	−0.104

Source: Authors' calculations from the Forbes 400.

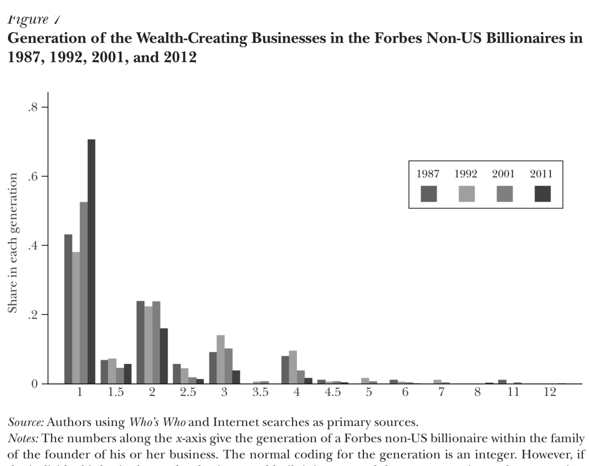
Categories of wealth-creating businesses behind the Forbes 400.

5.6 Figures 7 and 8: non-US billionaires

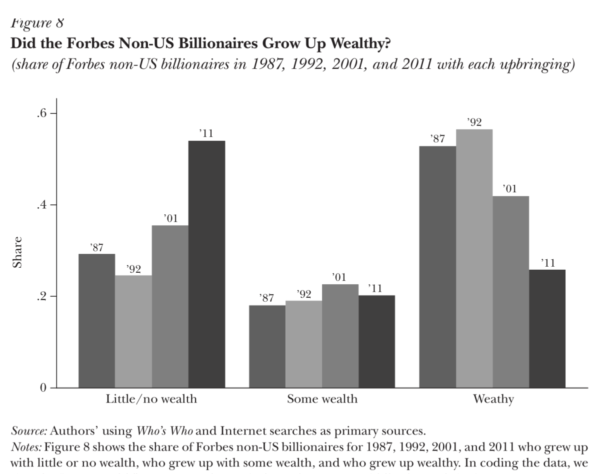
Read:

- Figure 7 shows a similar shift toward first-generation wealth creation among non-US billionaires.
- The first-generation share is high in 2011, while second-generation and later inherited fortunes decline.
- Figure 8 shows a large decline in the share who grew up wealthy.
- The little/no wealth category rises sharply among non-US billionaires by 2011.

Economic message: The reduced role of inherited wealth is not only a US phenomenon. The global billionaire data also show more first-generation fortunes.



(a) Generation of non-US billionaire businesses.



(b) Childhood wealth background of non-US billionaires.

5.7 Table 3: industries behind non-US billionaires

Read:

- For non-US billionaires, diversified/other remains the largest category in 2011 at about 30.4 percent.
- Retail/restaurant rises from 10.1 percent to 12.4 percent.
- Computer technology rises from 1.1 percent to 6.8 percent.

- Mining/metals rises from 1.1 percent to 7.0 percent.
- Construction falls from 9.0 percent to 3.4 percent.
- Finance categories grow, but less dramatically than in the US Forbes 400.

Economic message: Outside the United States, natural resources and diversified businesses play a larger role, but the global data still show growth in technology and first-generation wealth.

Table 3

Categories of the Wealth-Creating Businesses behind the Forbes Non-US Billionaires

(shares of the businesses of Forbes non-US billionaires)

	1987	*1992*	*2001*	*2011*	*Change 1987–2011*
Industrial					
Retail/Restaurant	0.101	0.101	0.109	0.124	+0.023
Technology – Computer	0.011	0.039	0.091	0.068	+0.056
Technology – Medical	0.045	0.050	0.042	0.039	−0.006
Consumer	0.157	0.128	0.113	0.098	−0.060
Media	0.067	0.067	0.053	0.030	−0.037
Diversified/Other	0.247	0.346	0.362	0.304	+0.057
Energy	0.022	0.022	0.034	0.049	+0.026
Construction	0.090	0.061	0.042	0.034	−0.056
Mining and metals	0.011	0.011	0.026	0.070	+0.059
Finance and investments					
Hedge funds	0.000	0.000	0.004	0.008	+0.008
Private equity/leveraged buyout	0.022	0.000	0.011	0.010	−0.012
Money management	0.034	0.061	0.072	0.065	+0.031
Venture capital	0.000	0.000	0.004	0.010	+0.010
Real estate	0.101	0.101	0.109	0.124	+0.023

Categories of wealth-creating businesses behind non-US billionaires.

6 Short summary

- The paper studies the rise in incomes and wealth at the very top.
- Its main point is that the rise is broad-based across occupations, not just CEOs of public companies.
- CEO pay rose substantially, but relative to other top 0.1 percent taxpayers it has been fairly stable since the late 1990s.
- Top hedge fund managers, private equity founders, law partners, and athletes also earned much more over time.
- The top 1 percent income share rose sharply in pre-tax market income, but the after-tax, after-transfer rise was more muted.
- The Forbes 400 became more first-generation and less inherited from 1982 to 2011.
- The share of Forbes 400 individuals who grew up wealthy fell, while those from some wealth rose.
- US Forbes wealth shifted toward technology, retail/restaurant, hedge funds, and private equity, and away from energy and real estate.
- Similar, though not identical, patterns appear among non-US billionaires.
- These facts are more consistent with scale, technology, and superstar returns than with a narrow CEO-power or social-norms explanation.

7 Conclusion

7.1 Core conclusion

Kaplan and Rauh argue that the rise in the return to top talent is a market-wide phenomenon. The evidence does not fit well with a story focused only on managerial power in public companies. Instead, top pay and top wealth rose across many settings where talent can now be applied over much larger markets.

7.2 Economic intuition

Information technology, communications, finance, and global markets increase the scale at which highly talented individuals can operate. A better CEO can manage a larger firm, a better hedge fund manager can deploy more capital, a better lawyer can work on larger transactions, and a better athlete can reach larger audiences. When scale rises, the return to being at the top rises.

7.3 Takeaway

The broad-based rise in top incomes should be interpreted as a combination of technological scale, market size, and skill-biased change. Managerial power, social norms, and taxes may matter, but the paper's patterns point most strongly to the market return to scalable top talent.